

# Shayan Shakeri

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## PROFESSIONAL SUMMARY

AI Research Engineer at BMO's AI Centre of Excellence. Built a multi-agent data layer and agent marketplace for the bank from the ground up, scaled them, and own them today; also built the eval pipeline that caught a production GenAI tool serving \$200B+ AUM downplaying investment risk. Earlier, took a proprietary subcutaneous neurostimulator from idea to production device at Epineuron. Co-founder of Merlyn Labs; open-sourced a flow-matching VLA integration for RLinf.

## CORE COMPETENCIES

Agentic Systems

Agent Evaluation

LLM Fine-tuning

Reinforcement Learning

Python & C/C++

PyTorch & JAX

## WORK EXPERIENCE

BMO AI Centre of Excellence

Sep 2025 - Present

AI Research Engineer

Toronto, Canada

- Built, scaled, and own BMO's multi-agent data layer and agent marketplace, from the ground up.
- At its core: a graph-based agentic system answering multi-hop relational queries across bank client data.
- Caught a production GenAI tool serving \$200B+ AUM systematically downplaying investment risk.
- Built the deterministic eval pipeline that found it: hundreds of synthesized inputs, run at scale.
- Designing evals for LLM agents that retrieve and reason over commercial banking and insurance policy.

Merlyn Labs

Aug 2025 - Present

AI Researcher & Co-Founder

Toronto, Canada

- Open-sourced a flow-matching VLA integration for RLinf, enabling RL on BEHAVIOR-1K in OmniGibson.
- Developing VLM judges that turn rollouts into dense, hard-to-game RL rewards.
- Showed  $\pi 0.5$ 's LIBERO-PRO failures are recipe-induced overfitting; proposed a recalibrated baseline.
- Conservative finetuning doubled position-swap success, 21% to 42%, with no architectural change.

Epineuron Technologies

May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op

Mississauga, Canada

- Owned a proprietary subcutaneous neurostimulator from idea to production device.
- Designed and assembled PCBs; wrote IEC 60601-1 / ISO 13485 validation protocols for clinical test runs.
- Part of the team behind PeriPulse, an FDA Breakthrough-designated device in multinational clinical trials.
- Extended battery life 900% through oscilloscope and power-analyzer testing.
- Modeled nerve-field penetration in COMSOL; the results set the electrode diameter.

## PROJECTS

Stanford BEHAVIOR-1K Challenge

8th Place, Standard Track

Identified proprioceptive collapse as a critical VLA failure mode (60% masking improved task success up to 48%); found chunked execution outperformed temporal ensembling 3x. Trained on 10,000+ demonstrations covering 22 of 50 scored task types; published a technical report and a LessWrong analysis.

BardSong

Closed Alpha

Built the end-to-end pipeline: Groq speech-to-text, Gemini scene extraction, image generation, narrated video. Fine-tuned an LLM on D&D sourcebooks for narrative coherence; in closed alpha with 23 dungeon masters.

## EDUCATION

M.Eng in AI & Robotics University of Toronto

Sep 2024 - Apr 2027 (anticipated)

B.Eng, Engineering Physics Co-op McMaster University

Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

## SKILLS

**Agents & ML:** Agents, Agent Evaluation, Alignment Evaluation, LLM Fine-tuning, Reinforcement Learning, Imitation Learning, Flow Matching  
**Systems & Languages:** Python, C/C++, PyTorch, JAX, OmniGibson, MuJoCo, ROS2, Sim-to-Real, PCB Design